

Report

My solution is developed in React and I have chosen to use JavaScript, Tailwind, CSS and a bit of TypeScript, to make the solution more maintainable and scalable. It is connected to a Web API that supports CRUD operations and interacts with a database consisting of three tables: TrumpThought, TrumpMerch and MerchCart.

Functionality:

The main functionality is derived from the Web API, which provides methods for CRUD operations. These methods are utilized via Http requests like GET, POST, PUT and DELETE from the frontend to manage the database.

Implementation:

The implementation of the TrumpVerse application is designed to ensure a seamless connection between the frontend and backend using efficient code. Axios is used across the project to handle all Http requests to the Web API, facilitating CRUD operations.

These operations are implemented through dedicated service modules for each feature, such as managing thoughts, merch and cart. This modular approach ensures that API interaction logic is centralized, reusable and easy to maintain.

Additionally, services are used to handle specific functionalities, including an image upload service written in TypeScript to ensure strong type-checking and improved maintainability for a feature that requires precise data handling, which allows users to upload images when creating new merch.

By organizing backend interactions into services, the code base remains clean and scalable. Axios also ensures smooth error-handling, enabling the application to provide clear feedback to users if an API call fail. This implementation is a way to make sure the application is efficient and user friendly.

Features:

The application offers four pages, each tailored to a specific functionality for the user. The Homepage is the introduction to the application where they get more information on how to use it. The Thought page allows the user to read, add and delete Trumps thoughts and quotes, offering a way to engage with the database. The Merch page showcases a collection of Trump merch, where the user can perform full CRUD operations like creating, updating, reading and deleting merch, as well as adding them to a shopping cart, where the circle at the cart SVG at the top updates automatically according to the total quantity of the cart.

The cart page provides a complete shopping experience, where users can adjust the quantity of the merch items in their cart, view a summary of the total quantity and cost of their selected items, and finalize their purchase with a button on the bottom of the page.

Changes made in the database are reflected instantly on the frontend. The integration between the Web API and frontend is designed with full functionality, with the database preloaded at least eight rows in both TrumpThought and TrumpMerch tables.

Universal design:

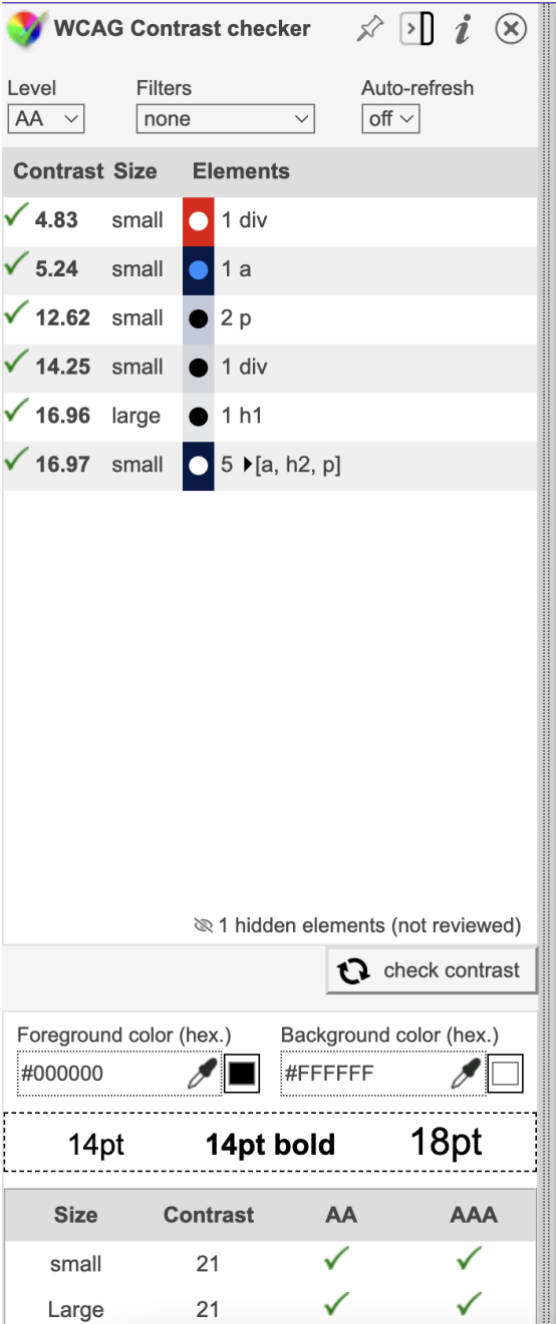
The application follows universal design principles to ensure it is accessible and user-friendly. Clear and concise error messages are displayed to users, directly addressing the problem to enhance usability and reduce confusion.

The color scheme and contrast follow the WCAG 2.1 guidelines, ensuring a contrast ratio of at least 4.5:1. the screenshots bellow is from WCAG Contrast Checker, from my pages in the application.

The layout is responsive, achieved through the use of CSS Grid, Flex and media queries, allowing the interface to adapt seamlessly across different devices and screen sizes.

All buttons, dropdowns, and input fields are equipped with proper labeling, making navigation intuitive and accessible for screen readers.

Additionally, interactive elements include feedback mechanisms such as hover effects, color changes, and click animations, which improves the clarity of actionable items and provide a more engaging user experience. These designs choices collectively ensure the application is inclusive and meets the needs of a diverse user base.

HomePage:

The screenshot shows the WCAG Contrast checker interface. At the top, there are controls for Level (AA), Filters (none), and Auto-refresh (off). Below this is a table of elements with columns for Contrast, Size, and Elements. The table lists six elements with their respective contrast ratios and sizes. At the bottom, there are color selection tools for foreground and background colors, and a table showing contrast ratios for different font sizes and weights.

Contrast	Size	Elements
✓ 4.83	small	1 div
✓ 5.24	small	1 a
✓ 12.62	small	2 p
✓ 14.25	small	1 div
✓ 16.96	large	1 h1
✓ 16.97	small	5 ▶[a, h2, p]

1 hidden elements (not reviewed)

check contrast

Foreground color (hex.) #000000 Background color (hex.) #FFFFFF

Size	Contrast	AA	AAA
small	21	✓	✓
Large	21	✓	✓

ThoughtPage:

 **WCAG Contrast checker**    

Level: Filters: Auto-refresh:

Contrast	Size	Elements
✓ 4.83	small	 1 div
✓ 5.24	small	 1 a
✓ 5.38	small	 1 button
✓ 6.47	small	 8 button
✓ 16.96	small	 46 ▶[h2, label, li, strong, small]
✓ 16.96	large	 1 h1
✓ 16.97	small	 5 ▶[a, h2, p]

1 hidden elements (not reviewed)

 check contrast

Foreground color (hex.)   Background color (hex.)  

14pt 14pt bold 18pt

Size	Contrast	AA	AAA
small	21	✓	✓
Large	21	✓	✓

MerchPage:

 **WCAG Contrast checker**    

Level: Filters: Auto-refresh:

Contrast	Size	Elements
✓ 4.83	small	2 ▶[div, button]
✓ 5.02	small	9 button
✓ 6.47	small	9 button
✓ 6.7	small	2 button
✓ 7.56	small	72 ▶[p, strong]
✓ 7.9	small	1 a
✓ 12.62	small	9 ▶[label, h3, h2]
✓ 16.96	small	3 h2
✓ 16.96	large	1 h1
✓ 16.97	small	5 ▶[a, h2, p]
✓ 21	small	18 ▶[div, h3]

🔍 2 hidden elements (not reviewed)

 check contrast

Foreground color (hex.)   Background color (hex.)  

Size	Contrast	AA	AAA
small	21	✓	✓
Large	21	✓	✓

CartPage:

 **WCAG Contrast checker**    

Level
AA ▾

Filters
none ▾

Auto-refresh
off ▾

Contrast	Size	Elements
✓ 4.83	small	 1 div
✓ 6.47	small	 2 button
✓ 6.7	small	 1 button
✓ 16.06	small	 12 ▶[div, p, strong, span]
✓ 16.06	large	 2 h3
✓ 16.96	small	 2 h2
✓ 16.96	large	 1 h1
✓ 16.97	small	 6 ▶[a, h2, p]
✓ 21	small	 4 button

 2 hidden elements (not reviewed)

 check contrast

Foreground color (hex.)
#000000  

Background color (hex.)
#FFFFFF  

14pt

14pt bold

18pt

Size	Contrast	AA	AAA
small	21	✓	✓
Large	21	✓	✓